NWT Module 4 Exam Answers

Transport Layer and Application Layer

Exact question wording from the supplied question bank. Two-mark answers use 4-5 sentence points or comparison rows; five-mark answers use 5-8 detailed points or comparison rows, with relevant examples and diagrams.

2 Marks | Q1. Examine the functions of the Application Layer and infer their role in providing network services to end users.

1. The Application Layer provides network services to applications used by end users.

2. It defines the message formats and communication rules used by applications such as browsers and email clients.

3. It supports services such as web access, file transfer, email, and name resolution through specific protocols.

4. It enables client and server applications to exchange requests and responses over the network.

5. For example, HTTP allows a browser to request a web page, while SMTP allows a mail system to send email.

2 Marks | Q2. Classify the different Application Layer protocols and select the appropriate protocol for a given network application.

1. Web-access protocols such as HTTP and HTTPS transfer web resources between browsers and servers.

2. Email protocols include SMTP for sending mail and POP3 or IMAP for retrieving or accessing messages.

3. FTP supports file transfer, making it suitable for uploading or downloading files between a client and server.

4. DNS resolves domain names to addresses, while DHCP automatically provides network configuration to hosts.

5. SNMP supports monitoring and management of network devices, so protocol selection depends on the application's purpose.

2 Marks | Q3. Compare connection-oriented and connectionless protocols, and justify their suitability for different communication scenarios.

Basis	Connection-oriented communication	Connectionless communication
Setup	It establishes a logical connection before normal data exchange.	It sends data without first establishing a transport connection.
Reliability	TCP provides acknowledgements and retransmission for reliable delivery.	UDP provides no built-in delivery or retransmission guarantee.
Ordering	TCP delivers application bytes in order.	UDP datagrams may arrive out of order or fail to arrive.
Overhead	Connection setup and reliability mechanisms add overhead.	The simpler mechanism reduces transport overhead.
Suitability	TCP suits file transfer and applications requiring ordered data.	UDP suits applications that prefer timely datagrams or small exchanges.

2 Marks | Q4. Assess the performance of TCP and UDP by comparing their advantages and limitations for different networking applications.

Basis	TCP	UDP
Advantage	It provides reliable, ordered byte-stream delivery using acknowledgements and retransmissions.	It provides simple datagram delivery with a small fixed header.
Control	It includes flow control and congestion control.	It does not include built-in flow control or congestion control.
Overhead	Its header is at least 20 bytes, and connection management adds overhead.	Its header is 8 bytes, and it requires no connection handshake.
Limitation	Retransmissions and ordered delivery can delay later application data.	The application must handle loss, duplication, or ordering when required.
Application	It is appropriate for FTP, mail transfer, and reliable application exchanges.	It is used for DNS queries and many real-time media applications.

2 Marks | Q5. Examine the working of the File Transfer Protocol (FTP) and infer its role in file transmission over a network.

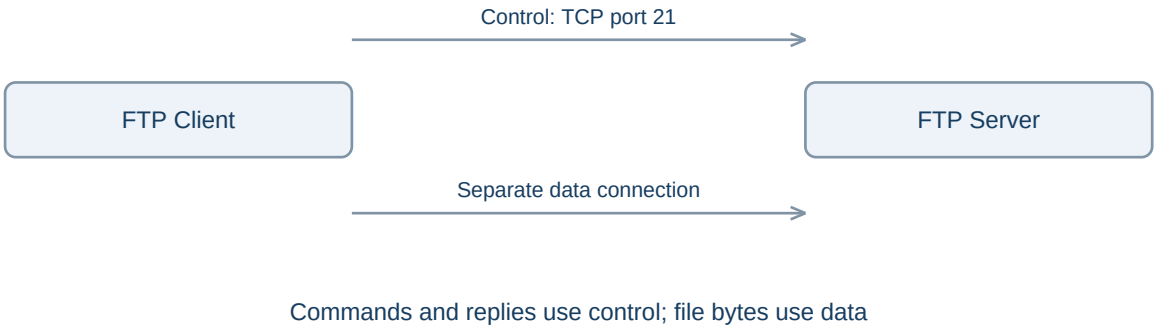
1. FTP stands for File Transfer Protocol and transfers files between a client and an FTP server.

2. It uses TCP and maintains a control connection, normally to server port 21, for commands and replies.

3. It uses a separate data connection for file contents and directory listings.

4. After authentication, a user can upload, download, list, or manage files according to the server's permissions.

5. Traditional FTP does not encrypt credentials or file contents, so protected alternatives are needed when confidentiality matters.



2 Marks | Q6. Analyze the relationship between the Transport Layer and the Application Layer, and infer how they cooperate to deliver end-to-end communication services.

1. The Application Layer defines the meaning, format, and exchange rules of application messages.

2. The Transport Layer provides process-to-process communication between applications on different hosts.

3. An application passes its messages to a transport protocol such as TCP or UDP for transmission.

4. At the receiver, port numbers help the transport layer deliver data to the intended application.

5. For example, an FTP application uses TCP's reliable byte-stream service to exchange commands and file data.

2 Marks | Q7. Illustrate the services provided by the Transport Layer and explain how they ensure reliable end-to-end communication.

1. The Transport Layer provides process-to-process delivery by using port numbers to identify applications.

2. It supports multiplexing and demultiplexing so multiple applications can communicate through a host's networking services.

3. TCP provides reliable delivery through sequence numbers, acknowledgements, and retransmission of missing data.

4. TCP flow control prevents a sender from overwhelming the receiver, while congestion control adjusts transmission to network conditions.

5. These services support reliable end-to-end communication when TCP is used, whereas UDP offers a simpler service without these guarantees.

2 Marks | Q8. Examine the role of a web browser and infer its significance in accessing web resources with a suitable example.

1. A web browser is a client application that requests, interprets, and displays web resources.

2. It uses a URL to identify a resource and normally relies on DNS to resolve the server's domain name.

3. It exchanges HTTP or HTTPS requests and responses with a web server.

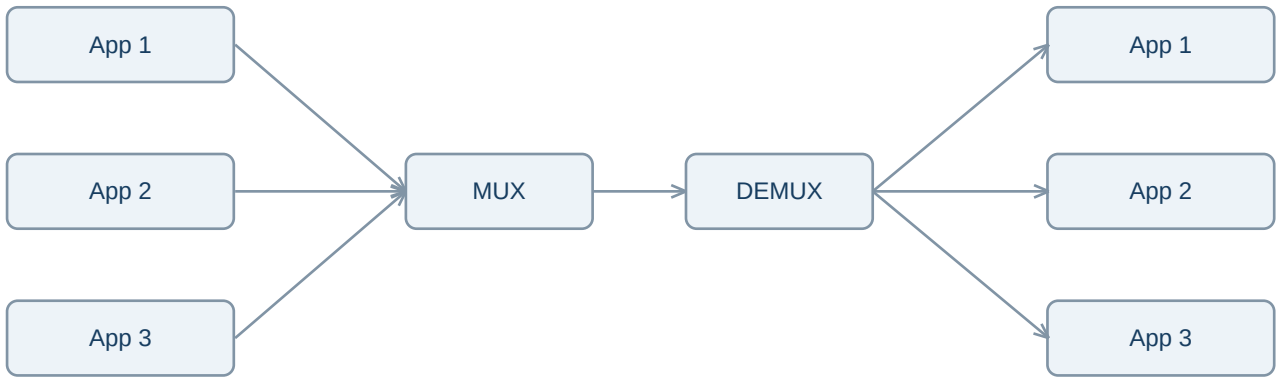
4. It processes HTML, applies CSS, and executes JavaScript to display an interactive web page.

5. For example, entering a college website's URL in a browser allows a student to view notices and access linked resources.

5 Marks | Q1. Distinguish between Multiplexing and Demultiplexing.

At the Transport Layer, multiplexing combines outgoing data from different application sockets for network delivery, while demultiplexing directs incoming data to the appropriate socket.

Basis	Multiplexing	Demultiplexing
Meaning	It accepts outgoing data from multiple application processes or sockets.	It delivers incoming transport data to the correct application process or socket.
Location	It occurs on the sending host for the data being transmitted.	It occurs on the receiving host for the data being delivered.
Operation	It adds transport headers and passes segments or datagrams to the network layer.	It examines transport information and selects the destination socket.
Identifiers	Source and destination ports identify the communicating application endpoints.	Ports and, where applicable, IP addresses identify the receiving socket.
Direction	It supports many applications sharing lower-layer network services.	It separates received traffic among the applications using those services.
Example	A browser and a mail application both send data through the same host network stack.	The receiver directs web traffic to a web socket and mail traffic to a mail socket.
Purpose	It supports efficient concurrent communication by multiple applications.	It prevents traffic from being delivered to the wrong application.



Many application streams share transport/network delivery

Illustration: Web and mail connections can share one network interface while remaining logically separate through their socket identifiers. Physical multiplexing such as FDM and TDM is related terminology at a different layer.

5 Marks | Q2. Simplify the principle of congestion control in detail.

1. **Meaning:** Congestion occurs when offered traffic exceeds the capacity of links or networking devices. Queues grow, delays increase, and packets may be dropped when buffers fill.

2. **Objective:** Congestion control adjusts the amount of traffic entering the network to avoid excessive overload. It differs from flow control, which protects an individual receiver from being overwhelmed.

3. **Congestion window:** TCP uses a congestion window, cwnd, to limit outstanding data according to network conditions. The sender is also constrained by the receiver's advertised window, so both network and receiver capacity affect sending.

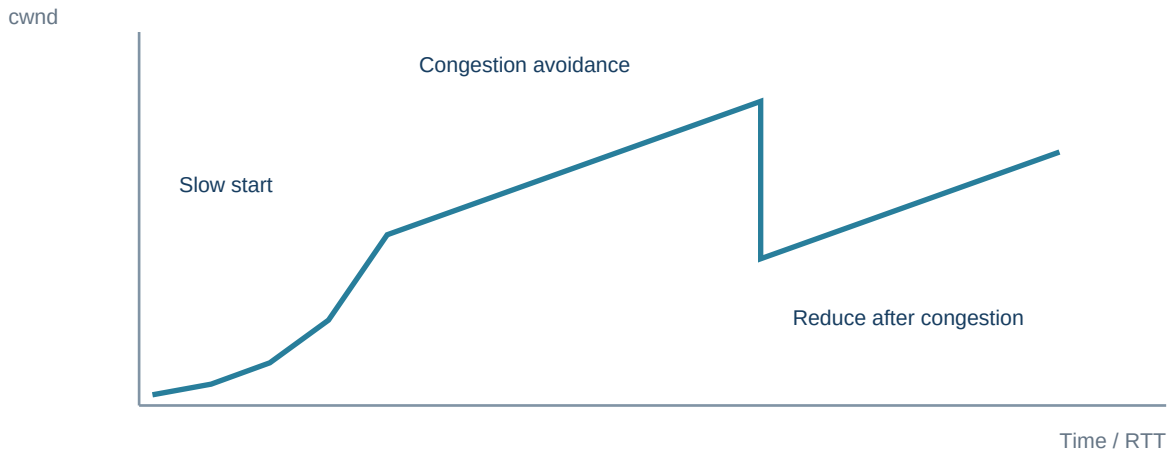
4. **Slow start:** TCP begins with a limited congestion window and increases it as acknowledgements arrive. During slow start, this growth is approximately exponential over successive round-trip times until a threshold or congestion event is reached.

5. **Congestion avoidance:** In the classic TCP model, growth becomes approximately linear after the slow-start threshold. This cautiously probes for additional network capacity rather than continuing rapid exponential growth.

6. **Congestion detection:** TCP can infer loss from a retransmission timeout or repeated acknowledgements indicating a missing segment. Explicit Congestion Notification can also signal congestion without requiring a packet drop.

7. **Response:** Classic TCP reduces its congestion window when congestion is detected and retransmits lost data as appropriate. A timeout generally causes a stronger reduction and a return to slow start, while fast retransmit and recovery respond to duplicate acknowledgements.

8. **Principle:** The additive-increase and multiplicative-decrease approach raises the sending rate cautiously and reduces it sharply after congestion. This helps balance utilisation and stability, although detailed behaviour varies between TCP algorithms.



Illustrative graph: The window rises rapidly during slow start, grows more gradually during congestion avoidance, and drops after a congestion signal. The curve is conceptual and is not measured experimental data.

5 Marks | Q3. Illustrate the Application Layer protocols in detail.

1. **Role:** Application Layer protocols define message formats, meanings, and communication rules between network applications. They support end-user services while relying on transport protocols for data delivery.

2. **HTTP and HTTPS:** HTTP exchanges web requests and responses between clients and servers. HTTPS protects HTTP communication using TLS.

3. **FTP:** File Transfer Protocol supports uploading, downloading, and managing files. It uses TCP with separate control and data connections.

4. **Email protocols:** SMTP submits and relays email, while POP3 retrieves messages and IMAP supports mailbox access and synchronisation. MIME extends email content formats to represent attachments and multiple media types.

5. **DNS:** Domain Name System maps domain names to records such as IP addresses. It allows applications to use readable names instead of remembering numerical addresses.

6. **DHCP:** Dynamic Host Configuration Protocol provides hosts with leased IP addresses and other network settings. It reduces the need to configure every device manually.

7. **SNMP:** Simple Network Management Protocol allows management systems to monitor and manage devices through agents. It supports reading managed values, modifying permitted values, and receiving event notifications.

Protocol	Service	Common port or transport
HTTP / HTTPS	They provide web-resource access.	TCP 80 / 443 are common; HTTP/3 uses QUIC over UDP.
FTP	It provides file transfer.	TCP 21 is the control port; data uses a separate connection.
SMTP	It provides mail submission and relay.	TCP 25 is used for relay; TCP 587 is common for submission.
POP3 / IMAP	They provide email retrieval or mailbox access.	TCP 110 / 143 are their conventional ports.
DNS	It provides name resolution.	UDP or TCP 53 is used.
DHCPv4	It provides automatic host configuration.	UDP 67 is the server port and UDP 68 is the client port.
SNMP	It provides device management.	UDP 161 is common for queries; UDP 162 for notifications.

5 Marks | Q4. Illustrate the Simple Mail Transfer Protocol (SMTP).

1. **Meaning:** SMTP stands for Simple Mail Transfer Protocol and is an Application Layer protocol for sending email. It handles message submission and transfer, while retrieval is handled by POP3 or IMAP.

2. **Components:** A sender's mail client submits a message to a mail server, which relays it towards the recipient's server. The recipient's mailbox stores the delivered message for later access.

3. **Transport:** SMTP operates over TCP to provide an ordered and reliable byte stream. Server-to-server relay commonly uses port 25, while authenticated submission commonly uses port 587.

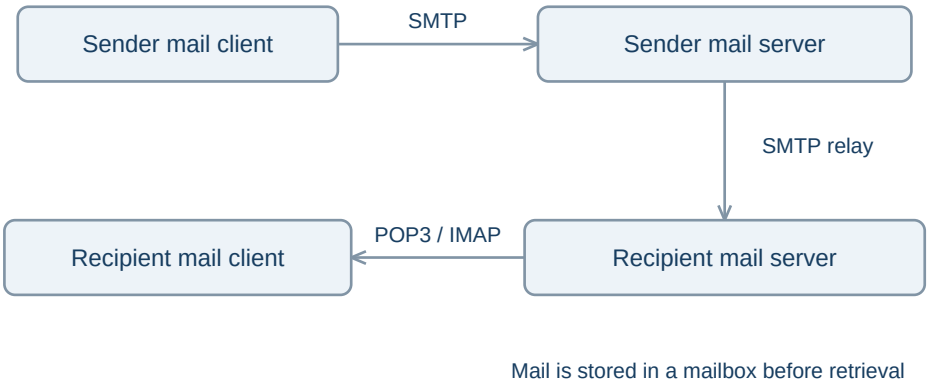
4. **Connection and greeting:** The SMTP client connects to the server and receives a greeting. It sends EHLO, or HELO for basic SMTP, to identify itself and begin the exchange.

5. **Envelope:** MAIL FROM specifies the envelope sender, and RCPT TO specifies an envelope recipient. These commands establish delivery information before the message content is transmitted.

6. **Message transfer:** DATA begins the transfer of message headers and body, followed by a line containing only a dot to mark completion. The server returns a status reply to indicate whether it accepted the message.

7. **Completion and relay:** QUIT closes the SMTP session, while a server may queue and retry messages that cannot be delivered immediately. Acceptance by one server does not necessarily mean final delivery to the recipient.

8. **MIME and protection:** MIME allows mail to carry attachments and different content types through suitable encoding. TLS can protect SMTP connections, but SMTP alone does not guarantee end-to-end encryption of stored message contents.




```
EHLO sender.example
MAIL FROM:<aayush@example.com>
RCPT TO:<student@example.org>
DATA
Subject: NWT Notes

Please find the revision notes.
.
QUIT
```

5 Marks | Q5. Elaborate the Dynamic Host Configuration Protocol (DHCP) in detail.

1. **Meaning:** DHCP automatically assigns IP configuration to hosts joining a network. It can provide an IP address, subnet mask, default gateway, and DNS-server information.

2. **Components:** A DHCP server manages an address pool, and a DHCP client requests configuration. A relay agent can forward DHCP messages between clients and a server on another network.

3. **DHCP Discover:** A client without an address broadcasts a DHCPDISCOVER message to find available servers. The message requests an offer of configuration.

4. **DHCP Offer:** A server replies with a DHCPOFFER containing an available address and configuration information. An offered address is not yet the client's confirmed lease.

5. **DHCP Request:** The client sends DHCPREQUEST to indicate its selected offer. During initial allocation, this also informs other servers that their offers were not selected.

6. **DHCP Acknowledgement:** The chosen server sends DHCPACK to confirm the lease and configuration. These initial steps are remembered as DORA: Discover, Offer, Request, and Acknowledge.

7. **Leases and ports:** DHCPv4 uses UDP port 67 at the server and UDP port 68 at the client. An address is leased for a period, and clients attempt renewal so that unused addresses can eventually return to the pool.

8. **Benefits and limitations:** DHCP reduces manual work and configuration errors and allows central management of network settings. A new client depends on an available server or relay, and rogue servers can provide incorrect settings.

Example: A laptop joining a college LAN may receive 192.168.1.25, subnet mask 255.255.255.0, gateway 192.168.1.1, and a DNS-server address from the DHCP server.

5 Marks | Q6. Compare MIME and POP3 protocols.

MIME and POP3 serve different parts of email handling: MIME describes message content, whereas POP3 retrieves messages from a mailbox.

Basis	MIME	POP3
Full form	MIME stands for Multipurpose Internet Mail Extensions.	POP3 stands for Post Office Protocol version 3.
Purpose	It extends email formats to represent different content types and attachments.	It retrieves email messages from a mail server to a client.
Nature	It is a set of message-format conventions and content headers.	It is an Application Layer protocol for client-server communication.
Operation	It describes content types, character sets, multipart boundaries, and transfer encodings.	It allows a client to authenticate, list messages, retrieve messages, and mark messages for deletion.
Data handling	It represents text, images, audio, and documents inside a message.	It retrieves stored messages regardless of whether their content uses MIME.
Transport and port	It has no dedicated transport connection or port of its own.	It normally uses TCP port 110, or port 995 for implicit TLS.
Example	Content-Type: application/pdf identifies a PDF attachment part.	The RETR command downloads a selected message from the mailbox.
Relationship	It makes an attachment understandable after the message is retrieved.	It can download a message containing MIME-formatted attachments.

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Example MIME headers for an attachment:
Content-Type: application/pdf
Content-Transfer-Encoding: base64
Content-Disposition: attachment; filename="notes.pdf"

Example POP3 retrieval commands:
USER student
PASS example-password
LIST
RETR 1
QUIT
```

Illustration: POP3 downloads an email containing NWT notes, and MIME tells the mail client that one part of the message is a PDF attachment. POP3 can leave messages on the server or delete them according to client policy and issued commands.

5 Marks | Q7. Elaborate the Simple Network Management Protocol (SNMP).

1. **Meaning:** SNMP stands for Simple Network Management Protocol and supports monitoring and management of network devices. It is commonly used for routers, switches, servers, and other managed equipment.

2. **Manager:** The SNMP manager is the management application that sends requests and receives responses or notifications. It can collect values such as interface counters to identify faults and performance trends.

3. **Agent:** An SNMP agent runs on a managed device and exposes management information. It responds to permitted requests and can report important events to the manager.

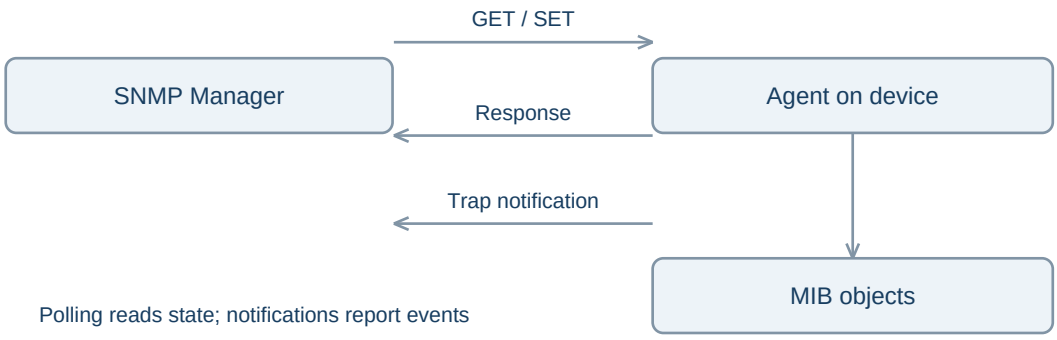
4. **MIB:** A Management Information Base defines managed objects organised in a hierarchy. Object identifiers, or OIDs, identify values such as interface traffic counters and device uptime.

5. **Reading information:** GetRequest reads specified values, while GetNextRequest traverses objects in order. GetBulkRequest, available in later SNMP versions, retrieves larger sets efficiently.

6. **Writing information:** SetRequest changes permitted managed values, subject to device support and access permissions. Response messages return requested information or indicate success and errors.

7. **Notifications and ports:** Trap notifications report events without acknowledgement, while InformRequest notifications request acknowledgement. UDP 161 is commonly used for manager-agent requests, and UDP 162 for notifications to managers.

8. **Security and use:** SNMPv1 and SNMPv2c use community strings, while SNMPv3 can provide authentication and encryption. For example, a manager can detect a failed interface through notifications and inspect traffic counters through polling.



Example: The manager periodically reads a switch's interface counters to monitor traffic, while the agent sends a link-down notification when an interface fails.

References: Supplied NWT question bank, Unit 4 notes, and NWT UNIT 4 notes.